MIL-P-28971A(MC)
8 May 1972
SUPERSEDING
MIL-P-28971(MC)
20 September 1971

MILITARY SPECIFICATION

PUMP, RECIPROCATING, POWER DRIVEN: 150 GALLONS PER MINUTE, DIAPHRAGM, GASOLINE-ENGINE-DRIVEN, SKID-MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type 4-inch, gasoline-engine-driven, skid mounted, reciprocating diaphragm pump.

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-601	-	Aluminum-Alloy Sand Castings.
QQ-I-652	-	Iron Castings, Gray.
QQ-S-741	-	Steel, Carbon, Structural Shape, Plate And Bar.
WW-N-351	-	Nipples, Pipe, Threaded.
ZZ-H-561	-	Hose, Rubber, And Hose Assemblies, Rubber Smooth Bore, Water.
PPP-B-621	-	Boxes, Wood, Nailed And Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.

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MILITARY

MIL-P-116	-	Preservation, Methods Of.
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	-	Plate, Identification, Instruction And Marking, Blank.
MIL-E-10062	-	Engines: Preparation For Shipment And Storage Of.
MIL-G-10924	-	Grease, Automotive And Artillery.
MIL-G-23827	-	Grease, Aircraft And Instrument, Gear And Actuator Screw.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129	-	Marking For Shipment And Storage.
MIL-STD-130	-	Identification Marking Of U. S. Military Property.
MS15003	-	Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS35932	-	Cock, Plug, Three-Way, Threaded Type.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D2000- Elastomeric Materials for Automotive Applications.
A242 - High-Strength, Low-Alloy Structural Steel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19102.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y., 10017.)

AMERICAN WELDING SOCIETY

Welding Handbook
Standard Qualification Procedure

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N.Y., 10017.)

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HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, N. Y., 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS

J492 Rivets and Riveting

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y., 10017.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 First article. The supplier shall furnish a first article for examination and test within the time frame specified (see 6.2) to prove, prior to starting production, that his production methods and choice of design detail will produce pumps which comply with the requirements of this specification. Examination and tests shall be as specified herein and shall be subject to surveillance and approval by the Government.

3.2 Parts and materials. Parts and materials shall be as specified herein. Parts and materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.2.1 Nonstandard parts and materials. When it has been determined that requirements cannot be met by parts and materials specified herein, immediate action shall be taken by the supplier to inform the contracting officer by submission of detailed data, requesting approval of nonstandard parts and materials.

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3.3 Design and construction. The pump shall be a 4-inch, double diaphragm type with a speed reduction system, a gasoline engine and all necessary accessories and connections. Easily accessible drain plugs shall be provided. The pump shall be skid mounted and in turn shall be mounted with all components on a Government furnished M762, 3/4-ton trailer (see 3.10). The pump shall be mounted in the trailer to allow connection of intake hose and discharge hose without undue or harmful bending of the hose. Components, particularly of the electrical, fuel and exhaust systems, shall be rust and corrosion resistant.

3.3.1 Pump. The pump shall be of the diaphragm lift and force type, shall be self-priming, and shall pump water containing solids. The pump shall be equipped with a suction passageway, suction valve with replaceable valve seat, suction-valve chamber, diaphragm chamber, flexible diaphragm discharge valve with replaceable valve seat, discharge valve chamber, and discharge passageway. Valves shall be of the ball type. Means shall be provided (without uncoupling the suction or discharge hoses) for removing obstructions which might prevent the valves from seating properly.

3.3.2 Pump case. The pump case shall be an iron casting conforming to QQ-I-652, class 20 or higher, steel having physical requirements in accordance with ASTM A242, or an aluminum casting conforming to QQ-S-601, alloy Almag 35, temper optional.

3.3.3 Diaphragm. The diaphragm shall be constructed of nylon-reinforced rubber conforming to ASTM D2000, 3BC620B14. The reinforcement shall consist of two layers of nylon cord which shall be treated for maximum adhesion to the rubber. The diaphragm shall be constructed to have all cord lengths equal and spiraled in opposite directions from the center outward to the outside diameter.

*3.3.4 Gasoline engine. The gasoline engine shall be 6-horsepower, 3,600 revolutions per minute, Briggs and Stratton Model 23A-R6, or equal.

3.3.5 Muffler. The muffler shall be positioned so that exhaust gases are directed away from the operator.

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*3.3.6 Speed-reduction system. Power from the engine shall be transmitted to the pump through a gear-train, roller-chain or a flexible spline drive enclosed in an oil-tight case. The rotating motion shall be changed to a reciprocating motion at the diaphragm by means of an eccentric crank, cam, or walking beam. The speed-reduction system shall drive the pump plunger and diaphragm at a normal operating rate of 65 strokes per minute. Bearings used in the speed-reduction system shall be either ball or roller type.

*3.3.7 Fuel system. The fuel system shall be provided with a three-way selector valve conforming to MS35932-1. The auxiliary connection of the selector valve shall be provided with a plug. A corrosion-resistant circular faceplate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK", AND "AUXILIARY." A fuel tank having a capacity of not less than 2 gallons shall be provided. The fuel tank shall be mounted in a manner to prevent fuel from spilling on the engine during refueling. A standpipe not less than 1/2 inch high from the bottom shall be incorporated in the fuel tank to prevent entry of dirt and foreign matter into the system.

*3.3.8 Suction and discharge ports. The suction and discharge ports shall be internally threaded with 4-inch NPT threads, and each port shall be provided with a zinc-coated, wrought iron or steel 4-inch pipe nipple, conforming to type I of WW-N-351, 6 inches long with 4-inch NPT threads.

*3.3.9 Hose. The pump shall be provided with four lengths of suction hose conforming to grade B, type I of ZZ-H-561. Each length of hose shall be 4 inches in diameter and 10 feet in length and shall be compatible to the nipples provided with the discharge and suction ports. The threads shall be 4-inch NPT.

3.3.10 Lubrication. The pump shall operate as specified herein when lubricated with military lubricants as follows. All surfaces requiring lubrication shall be provided with a means for lubricating.

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3.3.10.1 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun without the removal or adjustment of accessories or parts.

3.3.10.2 Enclosures with integral reservoir of lubricant. Enclosures such as gear cases and speed reduction housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for draining. The drain shall be fitted with a drain plug and shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the pump is in its normal operating position. Integral tubes or troughs may be used to convey the lubricant from the drain to the ground. The drain plug, the filling means, and the lubricant-level dipsticks or holes shall be accessible without disturbance of accessories or parts.

3.3.10.3 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall utilize greases as follows:

MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250°F.

MIL-G-10924 - For other bearings where operating temperatures do not exceed 175°F.

The pump shall be run-in (broken-in) and delivered with these greases. The bearings shall be cleaned before lubrication because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.3.10.4 Oil lubrication. Unless otherwise specified herein, all components shall be filled to the operating level with military oils designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

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3.3.11 Skid and trailer mounting. The skid shall be constructed of structural steel conforming to QQ-S-741. The skid shall be welded into a rigid support for the engine, pump and speed-reduction system. The skid-mounted pump shall be mounted on a standard Model M762, 3/4-ton trailer; the contractor shall modify the trailer for the most suitable mounting and the lowest possible center of gravity.

3.3.12 Steel fabrication. Steel used in the fabrication of the pump shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface shall not be acceptable. The forming of material shall be done by methods which will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All major bends shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.3.13 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed, and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.

Welding Qualifications of the ASME.

3.3.14 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

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3.3.15 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Bolt holes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

3.3.16 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member and shall be in accordance with SAE J492.

3.3.17 Threads. All screw threads shall be in accordance with H28.

3.3.18 Safety. All exposed parts subject to high operating temperatures or energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.3.19 Repair parts and tools. Unless otherwise specified (see 6.2), the contractor shall furnish a recommended listing of spare parts and tools required. No shipment of pumps shall be made unless repair parts and maintenance tools are included with the shipment or unless approval for shipment of the pump units without such items has been received from the contracting officer.

3.4 Maintainability. All assemblies shall be accessible for maintenance, repair, and replacement without the removal of other assemblies not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. All subassemblies shall be plainly marked for ease of assembly and for adjustment.

3.5 Performance. The pump shall be capable of the following performance characteristics.

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*3.5.1 Capacity. At a speed not exceeding 65 strokes per minute, the pump shall deliver not less than 150 gallons per minute at a static suction lift equivalent to 10 feet and 95 gallons per minute at a static suction lift equivalent to 20 feet. The pump shall deliver the rated capacity at 5,000 feet elevation.

3.5.2 Usage. The pump shall withstand hard usage, storage and operation in adverse conditions for extended periods of time, and operation when positioned 15 degrees from level.

*3.5.3 Standard conditions. The pump shall deliver, under standard conditions (atmospheric pressure of 29.92 inches of mercury and a water temperature of not less than 68°F) the capacities specified in 3.5.1. The pump shall not leak when subjected to a hydrostatic pressure of 50 pounds-force per square inch gage.

3.6 Dimensions. The overall dimensions of the skid-mounted pump shall enable pump to be mounted on the M762 3/4-ton trailer.

3.7 Weight. The total weight of the skid-mounted pump shall not exceed the maximum allowable weight for the M762 3/4-ton trailer.

3.8 Treatment and painting. The portions of the pump and its components and parts which are normally painted shall be cleaned, treated and painted in accordance with the acceptable commercial standards of industry. Color shall match Marine Corps green, No. 23, semigloss (see 6.3).

3.9 Marking.

3.9.1 Identification plates. The pump shall be identified in accordance with MIL-STD-130. Identification plates shall be photosensitive, anodized aluminum conforming to MIL-P-514.

3.9.2 Instruction plates. Each pump shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating or servicing the pump. Instruction plates shall be photosensitive, anodized aluminum conforming to MIL-P-514.

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3.9.3 Stenciling. The gross weight of the pump shall be stenciled on both sides of the pump in a position to be readily discernible. The letters shall be gothic capitals and the numerals shall be of similar appearance. The letters and numerals shall be 2 inches high where space permits.

3.10 Government furnished property. The Government shall furnish the contractor one M762, 3/4-ton trailer for each pump procured.

3.11 Workmanship. Workmanship shall be in accordance with the accepted commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

(c) Preparation for delivery (see 4.7).

4.3 First article inspection. The first article shall be examined for defects listed in 4.5. Presence of one or more defects shall be cause for rejection. The first article shall be subjected to tests in 4.6. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

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4.4.1 Individual. Each pump shall be examined for the critical defect specified in 4.5. Presence of a critical defect shall be cause for rejection.

4.4.2 Sample. Samples selected in accordance with MIL-STD-105 shall be examined for defects listed in 4.5. Acceptable quality level (AQL) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Defects. The pump shall be examined for the following defects:

Critical

1. Holes in casting.

Major

101. Dimensions not as specified.
102. Components missing or not as specified.
103. Identification or special marking missing or illegible.
104. Accessories, operating and safety instructions or technical manuals missing or not as specified.
105. Assembly not complete or not correct.
106. Safety devices or guards missing or not as specified.
107. Material not as specified.
108. Lubrication devices not as specified.

Minor

201. Treatment and painting not as specified.
202. Repair parts missing.

4.6 Tests. Prior to the following tests, the pump shall be lubricated with oils and greases as specified in 3.3.10. Oils shall be those designated for use in the ambient temperature at the place of test. Tests shall be performed in accordance with the applicable provisions of the test code for reciprocating pumps of the Standards of the Hydraulic Institute. Water shall be used as the test liquid, and all test data shall be corrected to standard sea level atmospheric conditions of 29.92 inches of mercury and to the specific weight of water (62.3 pounds per cubic foot) at a temperature of 68°F.

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4.6.1 Priming test. The priming test shall be conducted with the center of the pump inlet 25 feet above the level of the water. A 30-foot suction line of the same nominal size shall be connected to the pump inlet and the discharge shall be open to the atmosphere. It is permissible to first fill the pump body with water. The pump shall be operated at not more than 65 strokes per minute. Failure of the pump to prime itself within 5 minutes shall constitute failure of the test.

4.6.2 Hydrostatic test. The entire pump, including the suction passageway, suction-valve chamber, diaphragm chamber, discharge-valve chamber, and discharge passageway, shall be subjected to a hydrostatic pressure of 50 pounds-force per square inch for 10 minutes. The diaphragm, during the hydrostatic test, shall be in the position which gives maximum diaphragm chamber volume. Water leakage through any part of the pump shall constitute failure of the test.

4.6.3 Performance test. The pump shall be operated at 65 strokes per minute with static suction lifts equivalent to both 10 and 20 feet at standard conditions specified in 4.6. Failure of the pump to deliver 150 gallons per minute at the 10-foot suction lift conditions and 95 gallons per minute at the 20-foot suction lift conditions shall constitute failure of this test.

*4.6.4 Compatibility test. Each pump shall be operated continuously for 300 hours. A maximum shutdown time of 1 hour is allowable in any 24-hour period for servicing and minor adjustments. Diaphragm replacement is permissible only after the first 200 hours of test. The test shall be conducted with the pump operating at 65 strokes per minute, static suction lift equivalent to 10 feet and open discharge using a 4-inch suction hose. Failure of the pump to deliver 140 gallons per minute shall constitute failure of this test.

4.7 Inspection of preparation for delivery.

4.7.1 First article pack examination. When first article pack is required (see 5.1), it shall be examined for defects listed in 4.7.3. Presence of one or more defects shall be cause for rejection.

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4.7.2 Quality conformance inspection. For the purpose of this inspection, a completed pack prepared for shipment shall be considered a unit of product. Sampling for examination shall be in accordance with MIL-STD-105. Samples selected shall be examined for defects listed in 4.7.3. AQL shall be 2.5 percent defective. A lot shall consist of all completed packs of any one level offered for examination at one time.

4.7.3 Defects. The pack shall be examined for the following defects:

110. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
111. Disassembly not as specified for level A or B.
112. Surfaces of components requiring a contact preservative in accordance with the referenced document not coated for level A or B.
113. Engine and engine accessories not preserved and packaged in accordance with the referenced document.
114. Interior surfaces of the pump not coated with preservative as specified for level A or B.
115. Suction and discharge openings not sealed as specified for level A or B.
116. Maintenance tools not preserved and packaged in accordance with the referenced document for level A or B.
117. Consolidated packaging not as specified for level A or B.
118. All components not packed together as specified for level A or B.
119. Strapping not zinc coated for level A.
120. Depreservation guide not prepared and attached as specified herein.
121. Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY

5.1 First article pack. Unless otherwise specified (see 6.2), the supplier shall furnish a first article pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging,

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that the preservation, packaging, packing and marking comply with the preparation for delivery requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.

5.2 Preservation and packaging. Preservation and packaging shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly shall be the minimum necessary to accomplish reduction in cube in accordance with the manufacturer's recommendations; specifically the handle, the discharge valve body, and suction nipple shall be removed. Bolts, nuts, and washers removed shall be replaced in one of the mating parts and secured to prevent their loss, or packaged as specified in 5.2.1.6.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with, MIL-P-116.

5.2.1.3 Unprotected metal surfaces. Unpainted exterior metal surfaces of components of the pump, including surfaces exposed by disassembly, requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with type P-1 preservative.

5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with level A requirements of MIL-E-10062 as specified for type II classification.

5.2.1.5 Pumps. The interior surfaces of the casings and all parts within the casings shall be coated with preservative conforming to MIL-P-116, type P-10, grade 2. The suction and discharge opening shall be sealed with plastic or metal caps, plugs, or blank flanges, as applicable; or the openings shall be covered with paper conforming to MIL-B-121, type I, grade A, class 2 and the paper secured in place with tape conforming to PPP-T-60, type IV.

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5.2.1.6 Repair parts. Repair parts shall be preserved and packaged in accordance with criteria in MIL-P-116.

5.2.1.7 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with level A requirements of MIL-H-15424.

5.2.1.8 Consolidated packaging. Repair parts, maintenance tools, and loose small components shall be packaged in a close-fitting box conforming to PPP-B-636, type CF, class weather resistant. The contents shall be immobilized within the containers to prevent free movement and damage.

5.2.2 Level C.

5.2.2.1 Engine. The engine shall be preserved in accordance with the level C requirements of MIL-E-10062.

5.2.2.2 Repair parts. Repair parts shall be preserved and packaged as specified for level A.

5.2.2.3 Other components. Preservation and packaging of all other components of the pump shall be adequate to afford protection against deterioration and damage from the supplier to the initial destination. The supplier's standard practice may be used when it fills the above requirements.

5.2.3 Depreservation guide. A depreservation guide shall be prepared, placed in a waterproof envelope marked "Depreservation Guide" and secured in a conspicuous location on the pump. DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used.

5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Level A. Each complete pump shall be packed in a closefitting box conforming to PPP-B-621, class 2, style 7, with hood. The components shall be arranged, blocked, and braced within the box in a manner to secure minimum cube

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and prevent free movement and damage. Box closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall be zinc coated and placed adjacent to the skids.

5.3.2 Level B. Each complete pump, preserved and packaged as specified in 5.2, shall be packed as specified in 5.3.1 except boxes shall be domestic type and strapping shall not be required to be zinc coated.

5.3.3 Level C. Each complete pump, preserved and packaged as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.4 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pump is intended for use in dewatering applications, where the water bears sand, gravel or any other solids.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.

(b) First article time frame (see 3.1).

(c) When list of repair parts and maintenance tools is not required (see 3.3.19(.

(d) When a first article pack is not required (see 5.1).

(e) Time frame for first article pack (see 5.1).

(f) Applicable levels of preservation, packaging and packing required (see 5.2 and 5.3).

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6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-E-4527, DI-E-4528, DI-M-4704, DI-R-4803, DI-T-4902 and DI-T-4904 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps (Code CSY-10), Headquarters U.S. Marine Corps, Washington, D. C. 20380.

6.4 First article instructions. First article testing shall be in accordance with Section 4.

*6.5 Marginal notations. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - MC
Project No. 4320-N116

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